



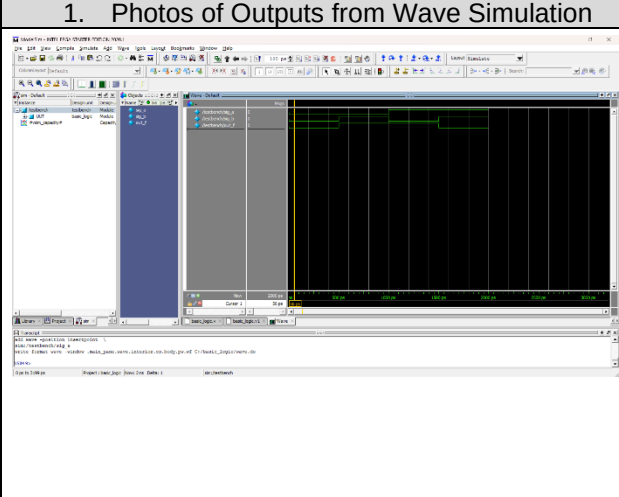
TESTBENCH WAVE SIMULATION USING MODELSIM

Name:	Lyka Sophia S. Jardeleza	Date:	04/21/2025
Program:	BSCpE	Section:	A361

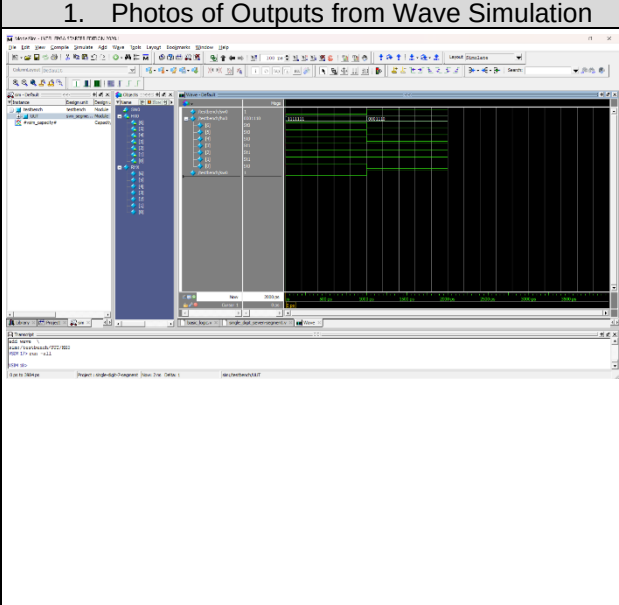
A. Guided Experiments:

(Please include the following: **1.** Photos of the simulation outputs and **2.** Photos of the codes that was inputted in the ModelSim software.)

Experiment 1: basic_logic Simulation

1. Photos of Outputs from Wave Simulation	2. Photos of Codes in ModelSim
	<pre>1 module testbench; 2 3 reg sig_a, sig_b; 4 wire out_f; 5 6 //instantiate logic circuit to be tested 7 basic_logic UUT(.a(sig_a), .b(sig_b), .f(out_f)); 8 9 10 initial 11 begin 12 sig_a = 0; sig_b = 0; # 500; //nano second 13 sig_a = 0; sig_b = 1; # 500; 14 sig_a = 1; sig_b = 0; # 500; 15 sig_a = 1; sig_b = 1; # 500; 16 end 17 endmodule 18 19 module basic_logic(a, b, f); 20 input a, b; 21 output f; 22 23 assign f = (~a&b) (a&~b); 24 endmodule</pre>

Experiment 2: Exercise #1 (Guided Exercise) – Single Digit 7-Segment

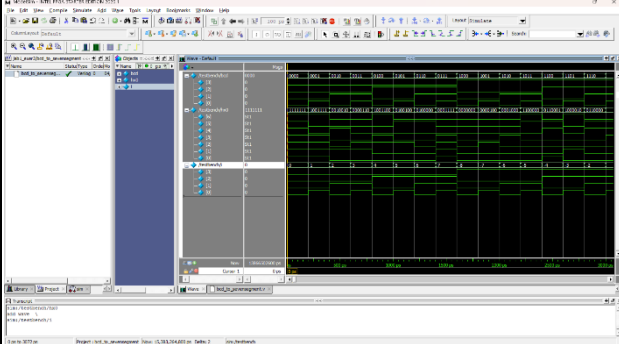
1. Photos of Outputs from Wave Simulation	2. Photos of Codes in ModelSim
	<pre>1 module testbench; 2 3 reg SW0; 4 wire [6:0] HX0; 5 6 //instantiate logic circuit to be tested 7 svn_segment_sim UUT(.SW0(SW0), .HX0(HX0)); 8 9 //test signals 10 initial 11 begin 12 SW0 = 0; # 1000; //nano second 13 SW0 = 1; # 1000; // nano second 14 end 15 endmodule 16 17 module svn_segment_sim(SW0, HX0); 18 input SW0; 19 output [6:0] HX0; 20 reg [6:0] RHX; 21 22 //combinational logic 23 always@(*) 24 begin 25 case(SW0) 26 0: RHX = 7'b1111_1111; //No display 27 1: RHX = 7'b000_1110; //F on 7-segment 28 endcase 29 end 30 31 assign HX0 = RHX; 32 33 endmodule</pre>

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B. Coding Exercise:

(Please include the following: **1.** Photos of the simulation outputs, **2.** Photos of the codes that was inputted in the ModelSim software, and **3.** The raw source code that you made for the coding exercise.)

Coding Exercise 1: Exercise #2 (Coding Exercise) – BCD to 7-Segment Decoder

1. Photos of Outputs from Wave Simulation	2. Photos of Codes in ModelSim
	<pre> 1 module testbench; 2 3 reg [3:0] bcd; 4 wire [6:0] hx0; 5 6 //instantiate logic circuit to be tested 7 bcd_svn_sim UUT(.BCD(bcd), .HX0(hx0)); 8 9 reg [3:0] i; 10 11 //test signals 12 initial 13 begin 14 for (i = 0; i < 16; i = i + 1) 15 begin 16 bcd = i; #200; 17 end 18 end 19 end 20 endmodule 21 22 module bcd_svn_sim(BCD, HX0); 23 24 input [3:0] BCD; 25 output [6:0] HX0; 26 reg [6:0] RHX; 27 28 //combinational logic 29 always@(*) 30 begin 31 case(BCD) 32 4'b0000: RHX = 7'b1111_1111; //0 33 4'b0001: RHX = 7'b1001_1111; //1 34 4'b0010: RHX = 7'b0011_0010; //2 35 4'b0011: RHX = 7'b0000_0110; //3 36 4'b0100: RHX = 7'b1001_1100; //4 37 4'b0101: RHX = 7'b0101_0100; //5 38 4'b0110: RHX = 7'b0101_0000; //6 39 4'b0111: RHX = 7'b0000_1111; //7 40 4'b1000: RHX = 7'b0000_0000; //8 41 4'b1001: RHX = 7'b0000_0100; //9 42 4'b1010: RHX = 7'b0000_1000; //A 43 4'b1011: RHX = 7'b1101_0000; //b 44 4'b1100: RHX = 7'b0101_0001; //C 45 4'b1101: RHX = 7'b1001_0010; //d 46 4'b1110: RHX = 7'b0101_0000; //E 47 4'b1111: RHX = 7'b0101_1000; //F 48 default: RHX = 7'b1111_1111; 49 endcase 50 end 51 52 assign HX0 = RHX; 53 54 endmodule </pre>

Raw Source Code (BDC to 7-Segment Decoder Module with Testbench):

```

module testbench;

    reg [3:0] bcd;
    wire [6:0] hx0;

    //instantiate logic circuit to be tested
    bcd_svn_sim UUT(.BCD(bcd), .HX0(hx0));

    reg [3:0] i;

    //test signals
    initial
        begin
            for (i = 0; i < 16; i = i + 1)
                begin
                    bcd = i; #200;
                end
            end
        end
    endmodule
                    
```



```
        end
    endmodule

    module bcd_svn_sim(BCD, HX0);

        input [3:0] BCD;
        output [6:0] HX0;
        reg [6:0] RHX;

        //combinational logic
        always@(*)
            begin
                case (BCD)
                    4'b0000: RHX = 7'b111_1111; //0
                    4'b0001: RHX = 7'b100_1111; //1
                    4'b0010: RHX = 7'b001_0010; //2
                    4'b0011: RHX = 7'b000_0110; //3
                    4'b0100: RHX = 7'b100_1100; //4
                    4'b0101: RHX = 7'b010_0100; //5
                    4'b0110: RHX = 7'b010_0000; //6
                    4'b0111: RHX = 7'b000_1111; //7
                    4'b1000: RHX = 7'b000_0000; //8
                    4'b1001: RHX = 7'b000_0100; //9
                    4'b1010: RHX = 7'b000_1000; //A
                    4'b1011: RHX = 7'b110_0000; //b
                    4'b1100: RHX = 7'b011_0001; //C
                    4'b1101: RHX = 7'b100_0010; //d
                    4'b1110: RHX = 7'b011_0000; //E
                    4'b1111: RHX = 7'b011_1000; //F
                    default: RHX = 7'b111_1111;
                endcase
            end

        assign HX0 = RHX;
    endmodule
```

C. Findings ,Observations, and Comments:

Guiding Questions :

1. What did you learn in the guided experiments? Did you learn something new?
2. What are your findings, observations, and comments in the guided experiments?
3. Did you solve the coding exercise?
4. Are there any difficulties during coding or set-up that you encountered?
5. What are your findings, observations, and comments in the coding exercise?

Throughout the guided experiments, I have learned how to apply the codes for the Single Digit 7-Segment in ModelSim and on how to use the functionalities of the app. The Coding Exercise was kind of bearable because I made the codes from the guided experiment a basis and I just added and changed some of the logic there. I was grateful because despite using the app for the first time, I was able to solve the coding exercise, and I had fun figuring out how to make the output waveform seen on the screen. All in all, the whole laboratory activity experience was challenging but fun. I'm glad that I got to learn new things from the ModelSim app.